

Investigation into a Self-Management App for People with Postural Tachycardia Syndrome (POTS)

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ABSTRACT

There is a need for research into the technology needs of people with Postural Tachycardia Syndrome (POTS). This paper examines the apps that are currently being used by people with POTS and explores what this population would like in an app designed specifically for people with this health condition. A prototype app is created that allows both the person with POTS and their carers to manage this condition. This prototype app is evaluated by potential users.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Postural Tachycardia Syndrome, self-tracking, self-management, carer, mobile apps

ACM Reference Format:

Jessica Mead. 2022. Investigation into a Self-Management App for People with Postural Tachycardia Syndrome (POTS). In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3491101.3516803>

1 INTRODUCTION

Postural Tachycardia Syndrome (POTS) is a chronic health condition which tends to affect women [1]. There is a lack of literature specifically discussing the technologies that people with POTS use, however, there is a need for this research in order to gain a greater understanding of the impact this health condition has on daily life and to allow the development of technologies that more fully support the needs of this population. The research reported in this paper meets this need by exploring the opinions of people with POTS regarding the apps they use and the features they would like to see in an app designed specifically for this population. The needs of carers are also considered.

The symptoms of POTS vary and can include fatigue, dizziness and brain fog [9]. Brain fog is defined as a “cognitive complaint

similar to mental fatigue” [10]. Symptoms of POTS can be aggravated by factors such as exercise and menstruation [7]. Monitoring heart rate, use of compression garments, graded exercise and consumption of salt and water can be helpful in reducing symptoms, however these strategies may not be helpful for every person, and some people with POTS benefit from social support, such as other people providing care [9].

Therefore, there is a need for an app that will help the person with POTS and their carers to perform useful strategies that will help manage the condition, while also having the option to track changes in health symptoms.

2 RELATED WORK

Apps can help people with health conditions achieve self-management of their condition [6, 12, 13]. Research into opinions on self-management technologies for Chronic Fatigue Syndrome (CFS) [2] found that available technologies were not suitable for people with CFS. For example, fitness trackers can be unsuitable as they are designed for people without a health condition, and so features such as encouraging the user to meet exercise targets could be inappropriate. This research is particularly relevant to people with POTS as both CFS and POTS share symptoms such as fatigue and exercise intolerance [7, 11].

Literature has explored technology-based collaboration when designing for people with chronic health conditions and their formal or informal carers in certain populations, such as an app for older people with cognitive impairments and their carers [8]. Literature has also examined apps that a parent can use to monitor the health condition of their child, such as the evaluation of an app that helps adolescents and parents to understand and manage the adolescent’s type 1 diabetes [4] and research into parent and clinician preferences for an adolescent asthma app [3]. There is a need for further research examining technologies that support collaboration between a person and their carer in other groups, such as adults that suffer from chronic fatigue [2]. This is because there is a need for technologies that allow carers to better understand the health condition of the person they provide care for and that help carers to assist the person in achieving self-management of their health condition.

3 RESEARCH OVERVIEW

3.1 Participants

Two online surveys are reported in this paper with participants aged 18 or over. Surveys 1 and 2 were open both to people who have a formal diagnosis of POTS and people who believe they have POTS

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3516803>

but do not have a formal diagnosis. This is because people may self-diagnose with POTS [5] and an app for this condition needs to be relevant to those who cannot get a diagnosis formally recognized but would still benefit from an app for symptom management. Survey 2 also recruited a further group: friends and family members of a person with POTS.

84 people responded to Survey 1. Of these respondents, 81 are female, and one did not give their gender. 75 respondents have a diagnosis of POTS and nine have POTS symptoms but no formal diagnosis. Survey 2 had 15 responses. 10 of these respondents have a diagnosis of POTS, two respondents have symptoms of POTS but no diagnosis and three respondents are the friend or family member of a person with POTS. Two friends or family members provide care for the person with POTS. Some respondents did not provide an answer for every question.

3.2 Data Collection

Online surveys were chosen as the method of data collection for this research as it enabled respondents to take part in their home at their own pace. All participants provided informed consent. A full review of the ethical implications of the research was conducted before the surveys took place with approval granted by the University of Bath Research Ethics Approval Committee for Health. No financial incentives were offered to respondents. Both surveys were distributed through a UK POTS Facebook support group, which is a private group the researcher joined. Survey 1 was also distributed by the POTS UK charity on their Facebook page and website. Respondents for Survey 1 were asked if they would like to be contacted to take part in Survey 2.

4 SURVEY 1

Survey 1 consisted of qualitative and quantitative questions exploring what apps respondents use and what respondents want from an app designed specifically for people with this health condition. An inductive approach was taken to identify the key themes in the qualitative data. The main findings are reported below.

4.1 Survey 1 Findings

A list of POTS symptoms was compiled through a literature review [5, 9, 10]. Respondents selected the symptoms they would most like a POTS self-management app to support. The results are shown in Figure 1. 80 respondents selected three or more symptoms.

Respondents were asked how many apps they currently use to manage their health. 50% do not use an app, 26% use one app, and 24% use multiple apps. Respondents were asked which apps they use. The results are shown in Figure 2 below. If respondents use four or more apps, they gave their favorite three apps.

Respondents were asked to select the POTS specific goals they want an app to help them to achieve. 73 respondents selected drinking three liters of water, 50 respondents selected eating more salt, and 27 respondents selected wearing compression garments. Respondents were asked to select additional features they want the app to provide. 77 respondents selected the ability to track POTS symptoms and 56 respondents selected information on POTS and how it can be managed.

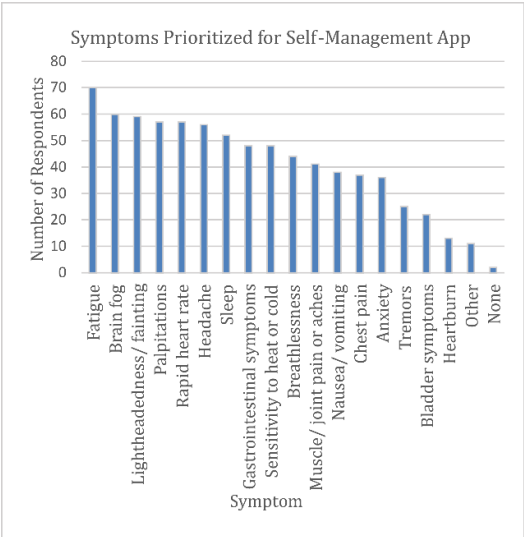


Figure 1: The symptoms respondents most want an app to support.

Open questions were asked to understand the respondents’ opinions on the apps they use and what they would like in an app designed specifically for POTS. The key themes are given in Table 1, Table 2 and Table 3.

5 PROTOTYPE APP

A prototype app was created based on the results of Survey 1. The findings from Survey 1 suggest that an app for people with POTS needs to include a range of functionality that allows the person to collect data that empowers them to understand their health condition. This is achieved by allowing the user to track symptoms alongside daily activities, goals and medications. Additional features that were discussed in Survey 1 are also included in the prototype app, such as the app allows the user to quickly track how they feel, the app has a place to enter notes on medication and medical appointments, the app provides information on POTS and management tips and the app allows other people to view tracking results, medication taken and other notes and allows other people to assist the person with POTS by entering their activities or symptoms on their behalf.

6 SURVEY 2

A second survey asked participants for their opinion on the prototype app. In particular, the survey collected respondents’ opinions on the feature of the app that allows another person to use the app alongside the person with POTS. Two videos of the app were shown to the participants. The first video shows the user tracking their salt consumption. The second video shows the user tracking the symptom brain fog. Both videos also describe the other features of the app as described in Section 5. The respondents were told that the ability to track the menstrual cycle alongside symptoms would be added to the app, along with reminders to perform tasks such as drinking water. The main findings are reported below.

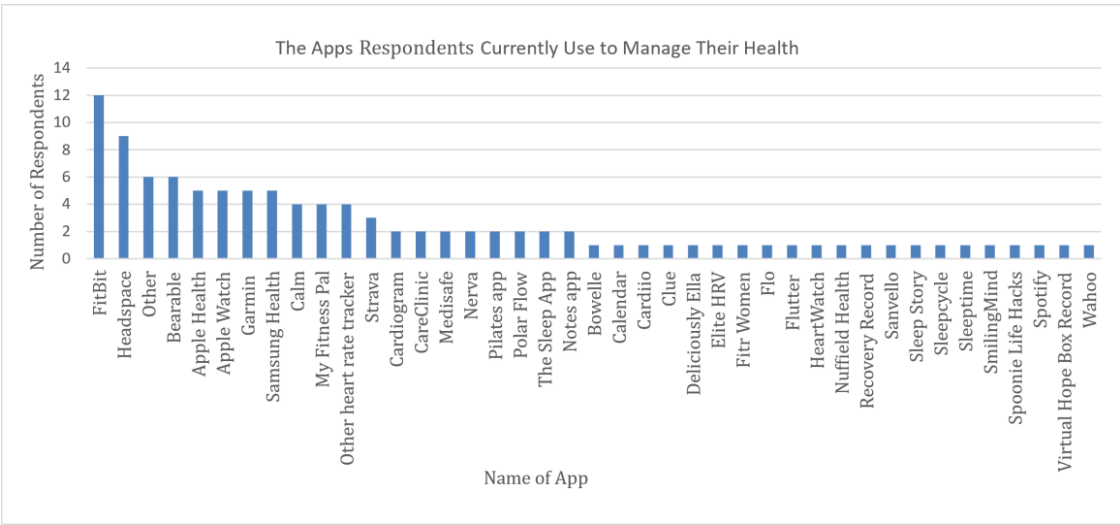


Figure 2: The apps survey respondents use to manage their health.

Table 1: What respondents like or find helpful about the apps they currently use to manage their health.

Number of Respondents	Key Theme	Additional Details Relating to the Theme
20	The app enables heart rate tracking	Seven respondents reported liking the heart rate tracking of their FitBit.
9	The app enables diet tracking	Some respondents track their diet alongside health symptoms to investigate correlation between the two.
9	The app helps reduce anxiety or stress	
9	The app helps improve or monitor sleep	
8	The app identifies symptom triggers	
6	The app tracks activity levels	The benefit of this was often mentioned as enabling pacing to reduce symptoms worsening.
5	The app shows long term trends	
4	The app tracks menstrual cycle details	

Table 2: What respondents do not like about apps they use or have used in the past.

Number of Respondents	Key Theme	Additional Details Relating to the Theme
15	The app cannot track everything required	
12	The app is too complicated or time consuming	3 people said Bearable was too complicated.
10	The app is too expensive	The most discussed were meditation apps.
9	The app is not accurate enough	
7	The app is not designed for chronic illness	

6.1 Survey 2 Findings

Some respondents suggested improvements to the prototype app. Three respondents said sharing data with their doctor would be useful. One respondent said it would be useful to track symptoms of other health conditions.

Twelve respondents thought adding menstrual tracking alongside symptom tracking to the app would be useful, two thought it would not be useful and one was not sure.

6.1.1 Respondents Who Have Someone Providing Care for Them or Who Provide Care for a Person with POTS. The seven respondents who have POTS and receive support from another person were asked for their opinions on an app that another person could use to help track and manage their condition. Three respondents thought this would be useful as it might help other people to understand how they are feeling and when they need help. Two respondents were not sure about this feature and one person explained they

Table 3: Features respondents would like to see in a POTS specific app (in addition to the features selected previously).

Number of Respondents	Key Theme	Additional Details Relating to the Theme
16	Track or set goals for exercise	Some respondents want an app to either prevent them from over-exercising or encourage them to perform suitable types of exercise.
11	Track patterns in symptoms	
10	Track heart rate	Some respondents said they want to see correlations between certain activities and their heart rate and symptoms during this activity.
10	Provide physical exercises such as physiotherapy	
8	Record information about medical appointments	This is needed to stop symptom worsening.
8	Provide support with pacing	
6	Allow socialising with other people with POTS	
6	Link to wearable technology for automatic tracking	
5	Manage medications	Medication reminders were also requested.
5	Export data from app	This was often to show the person's doctor.
3	Track symptoms alongside menstrual cycle	

anticipated their carer not engaging with the app. Four respondents thought sharing their lifestyle changes, medication tracking data or goal setting with another person would be useful. One participant explained that it would be useful for their carer to be able to see what medication they had taken. Three respondents were not sure if this feature would be useful as they reported being more organized than their carer or they thought having two people entering data could be confusing. When respondents were asked what they thought of the feature that allowed them to share their symptom tracking data with another person, all seven respondents indicated this would be helpful. One respondent explained that it would be useful for their carer to understand when they are experiencing particularly bad symptoms. Three respondents thought the feature that allows others to enter data into the app on their behalf would be useful, with two respondents explaining that their carer could track when they are too tired to track themselves. Three were not sure and one said this feature was not useful as the carer may forget to enter data. One participant said they would not want another person to enter or alter data unless they received consent every time. Both respondents who provide care thought they would use the features that enables them to see the goals and view and track the symptoms of the person with POTS.

6.1.2 Respondents with POTS Who Do Not Have Someone Providing Care. The five respondents who do not have someone looking after them were asked about sharing their lifestyle changes, medication tracking or goal setting data. All five respondents thought this would be useful. Two respondents said it would help other people understand their daily lives. Four respondents thought sharing symptom tracking data would be useful and one respondent thought it would not be useful.

6.1.3 Impact of Results of Survey 2 on the Design of an App for POTS. The results of Survey 2 show that sharing data from a self-management app with friends or family could help the person with POTS to explain symptoms that would otherwise be invisible. However, the responses suggest that not every carer would engage with

such an app. Furthermore, there needs to be careful consideration of how the app can be designed to allow multiple users to add or edit the same data.

7 CONTRIBUTION

This paper identifies a need for new technologies that support people with POTS and provides original research on the technology needs of people with POTS that could have implications for the designers of such technologies. This paper also contributes an extension of the currently available research examining the importance of designing apps for people with chronic health conditions that can also be used by a caregiver.

ACKNOWLEDGMENTS

I would like to thank the participants who took part in the surveys and my supervisor for his guidance.

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